

CHAPTER 1: MACAULAY'S METHOD

[William Herrick Macaulay © 1919]

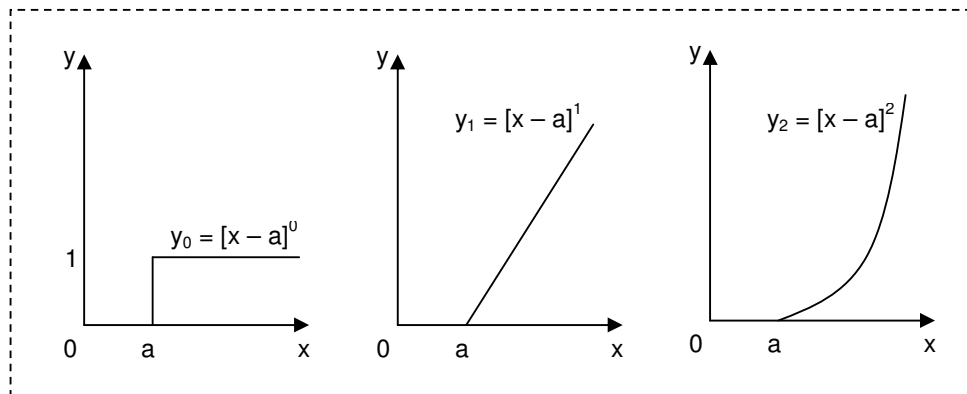
1.1 INTRODUCTION

The direct integration method becomes very tedious, too involving and time consuming if the BMD is piece-wise continuous (see example 3 of chapter 0). In each region there will be two constants of integration to be solved for. This can lead to a large number of simultaneous equations to be solved in order to obtain the constants of integration.

Macaulay's method is a modification of the direct integration method, based on the characteristics of **singularity functions**, which enables us to overcome the above difficulty.

1.2 SINGULARITY FUNCTIONS

Consider the mathematical functions, called singularity functions, shown below.



The singularity functions are expressed by use of square brackets, thus:

$$y_n = [x - a]^n \quad \text{for} \quad n = 0, 1, 2, 3, \dots$$

Singularity functions have three characteristics: -

1. $y_n = [x - a]^n = 0$ for all $x \leq a$

The function is **non-existent** for all values of x equal to or less than a

2. $y_n = (x - a)^n$ for $x > a$

The function takes its **actual mathematical value** for all values of x greater than a .

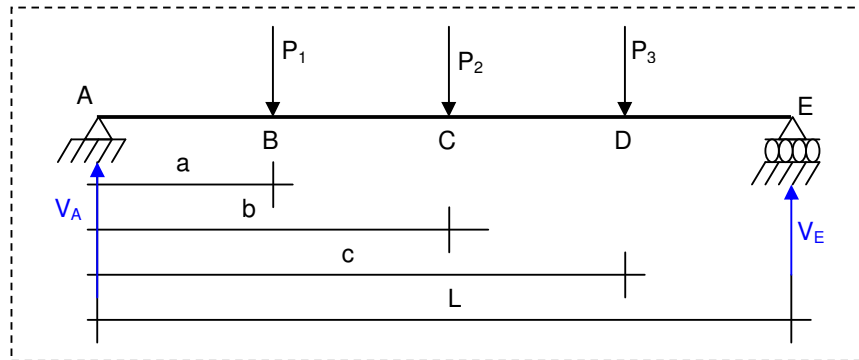
[Once the function starts, it continues to the “end” of x – axis]

3. $\int [x - a]^n dx = [x - a]^{n+1}/n+1 + C$ for all $n \geq 0$

The square bracket is treated as the variable during integration.

1.3 MACAULAY'S MOMENT EXPRESSION

Consider a simply supported beam subjected to point loads as shown below.



The bending moment expression in the various regions is conventionally expressed as: -

Region AB: $0 \leq x \leq a$: $M = V_A x$

Region BC: $a \leq x \leq b$: $M = V_A x - P_1(x - a)$

Region CD: $b \leq x \leq c$: $M = V_A x - P_1(x - a) - P_2(x - b)$

Region DE: $c \leq x \leq L$: $M = V_A x - P_1(x - a) - P_2(x - b) - P_3(x - c)$

According to Macaulay, a single bending moment expression for the whole beam may be written, using **square brackets**, as:

$$M = V_A x - P_1[x - a] - P_2[x - b] - P_3[x - c]$$

Note: 1. In the above expression, all the square brackets are to power 1.

[The bending moment diagram is piece-wise linear].

2. The expression is valid over the whole beam.

[Because characteristics number 1 and 2 of singularity functions apply].

1.4 SOLUTION PROCEDURE

The solution procedure may be outlined as follows:

1. Determine support reaction components to obtain a free body diagram.
2. Use square brackets to express the bending moment in the **last region** of the beam.
3. Integrate twice, observing characteristic number 3.
4. Apply boundary conditions, observing characteristic no. 1 and 2; solve for C₁ and C₂.
5. Obtain mathematical expressions for the deflection and slope profiles.
6. Use these expressions to evaluate the required displacements.
7. Sketch the D-shape.

Worked Examples

Example 1: Determine δ_D

Assume $EI = \text{constant}$

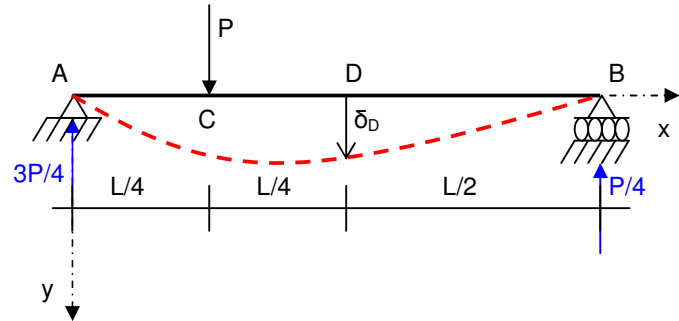
Solution

$$M = 3Px/4 - P[x - L/4]$$

$$EI \frac{d^2y}{dx^2} = -3Px/4 + P[x - L/4]$$

$$EI \frac{dy}{dx} = -3Px^2/8 + P[x - L/4]^2/2 + C_1$$

$$EI y = -3Px^3/24 + P[x - L/4]^3/6 + C_1x + C_2$$



Boundary conditions

$$1. y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. y|_L = 0 \Rightarrow -3PL^3/24 + P(3L/4)^3/6 + C_1L = 0 \Rightarrow C_1 = 7PL^2/128$$

Deflection profile: **$EI \delta = -Px^3/8 + P[x - L/4]^3/6 + 7PL^2x/128$**

At midspan, $x = L/2$: $EI \delta_D = -PL^3/64 + P(L/4)^3/6 + 7PL^2(L/2)/128 = 11PL^3/768$

$$\Rightarrow \delta_D = 11PL^3/768EI$$

Example 2: Determine δ_D .

Assume $EI = \text{constant}$

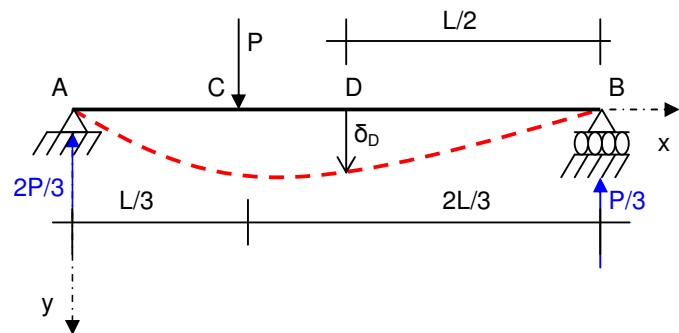
Solution

$$M = 2Px/3 - P[x - L/3]$$

$$EI \frac{d^2y}{dx^2} = -2Px/3 + P[x - L/3]$$

$$EI \frac{dy}{dx} = -2Px^2/6 + P[x - L/3]^2/2 + C_1$$

$$EI y = -2Px^3/18 + P[x - L/3]^3/6 + C_1x + C_2$$



Boundary conditions

$$1. y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. y|_L = 0 \Rightarrow -2PL^3/18 + P(2L/3)^3/6 + C_1L = 0 \Rightarrow C_1 = 5PL^2/81$$

Deflection profile: **$EI \delta = -Px^3/9 + P[x - L/3]^3/6 + 5PL^2x/81$**

At midspan, $x = L/2$: $EI \delta_D = -PL^3/72 + P(L/6)^3/6 + 5PL^2(L/2)/81 = 23PL^3/1296$

$$\Rightarrow \delta_D = 23PL^3/1296EI$$

Example 3: Determine δ_C . Assume $EI = \text{constant}$

Solution

$$M = Px/2 - P[x - L/2]$$

$$EI \frac{d^2y}{dx^2} = -Px/2 + P[x - L/2]$$

$$EI \frac{dy}{dx} = -Px^2/4 + P[x - L/2]^2/2 + C_1$$

$$EI y = -Px^3/12 + P[x - L/2]^3/6 + C_1x + C_2$$

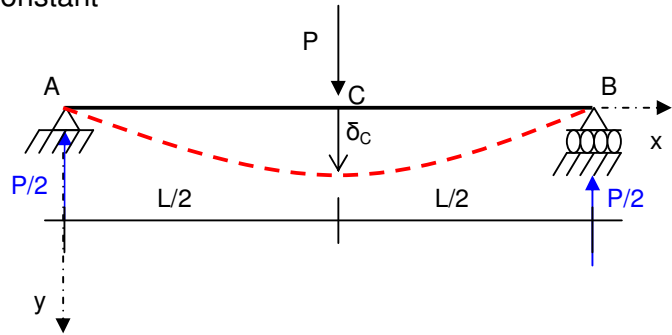
Boundary conditions

$$1. y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. y|_L = 0 \Rightarrow -PL^3/12 + P(L/2)^3/6 + C_1L = 0 \Rightarrow C_1 = PL^2/16$$

$$\text{Deflection profile: } EI \delta = -Px^3/12 + P[x - L/2]^3/6 + PL^2x/16$$

$$\text{At midspan, } x = L/2: EI \delta_C = -PL^3/96 + 0 + PL^2(L/2)/16 = PL^3/48 \Rightarrow \delta_C = PL^3/48EI$$



Example 4

Determine δ_C and $\delta_{\max}$ within span AB

stating where it occurs. Assume $EI = \text{constant}$.

Solution

$$M = 24x - 24x^2/2 + 252[x - 6]$$

$$EI \frac{d^2y}{dx^2} = -24x + 24x^2/2 - 252[x - 6]$$

$$EI \frac{dy}{dx} = -24x^2/2 + 24x^3/6 - 252[x - 6]^2/2 + C_1$$

$$EI y = -24x^3/6 + 24x^4/24 - 252[x - 6]^3/6 + C_1x + C_2$$

Boundary conditions

$$1. y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. y|_6 = 0 \Rightarrow -24(6)^3/6 + 24(6)^4/24 + C_1(6) = 0 \\ \Rightarrow C_1 = -72$$

Deflection profile

$$EI \delta = -4x^3 + x^4 - 42[x - 6]^3 - 72x$$

At the free end C: $x = 9$

$$EI \delta_C = -4(9)^3 + (9)^4 - 42(3)^3 - 72(9) = 1863$$

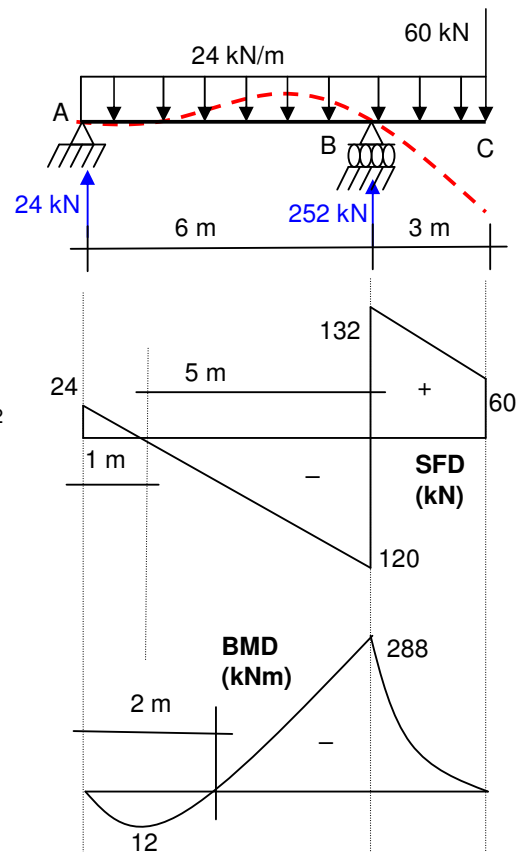
$$\Rightarrow \delta_C = 1863/EI$$

Within span AB, max deflection occurs when the slope = 0

$$EI \frac{dy}{dx} = -24x^2/2 + 24x^3/6 - 252[x - 6]^2/2 - 72 = 0 \Rightarrow -12x^2 + 4x^3 - 0 - 72 = 0$$

Solving (by trial and error), $x \approx 4.081\text{m}$

$$\text{Hence } \delta_{\max} \approx \{-4(4.081)^3 + (4.081)^4 - 72(4.081)\}/EI \Rightarrow \delta_{\max} \approx -288.33/EI$$



1.5 SOME SPECIAL CIRCUMSTANCES

In certain circumstances, Macaulay's method may appear to be handicapped. In such cases, it requires some combination of mathematical and engineering ingenuity to make the method work. Examples of such cases are discussed below.

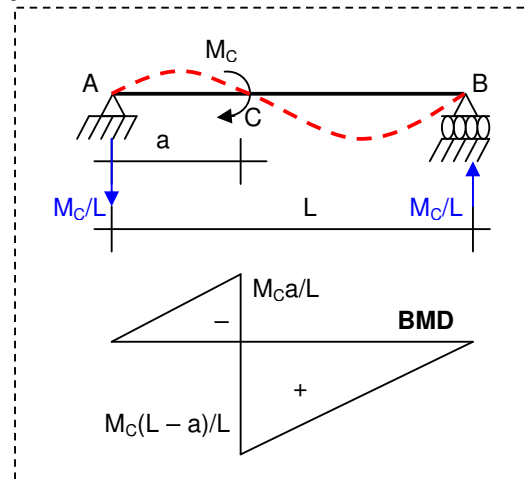
1.5.1 Concentrated Moment

Consider a simply supported beam subjected to a concentrated bending moment as shown below. The support reactions at A and B are calculated using equilibrium as usual. Note that the reactions are independent of the dimension a

Considering region CB, $a \leq x \leq L$, the bending Moment expression for the whole beam is given by: $M = -M_C x/L + M_C$

But this expression shows that at A, when $x = 0$ the bending moment has the value M_C !

This is NOT true.



In order to overcome this anomaly the bending moment expression for the whole beam should be written as: $M = -M_C x/L + M_C [x - a]^0$

It is now clear that in region AC, $M = -M_C x/L$, while in region CB, $M = -M_C x/L + M_C$

Example 5

Evaluate δ_D and δ_E . Assume $EI = \text{constant}$.

Solution

$$M = -M_C x/L + M_C [x - L/2]^0$$

$$EI \, d^2 y/dx^2 = M_C x/L - M_C [x - L/2]^0$$

$$EI \, dy/dx = M_C x^2/2L - M_C [x - L/2]^1 + C_1$$

$$EI \, y = M_C x^3/6L - M_C [x - L/2]^2/2 + C_1 x + C_2$$

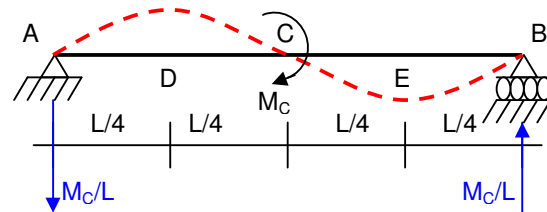
$$\text{Boundary conditions: } 1. \, y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. \, y|_L = 0 \Rightarrow M_C L^3/6L - M_C (L/2)^2/2 + C_1 L = 0 \Rightarrow C_1 = -M_C L/24$$

$$\text{Deflection profile: } EI \, \delta = M_C x^3/6L - M_C [x - L/2]^2/2 - M_C Lx/24$$

$$\text{At D, } x = L/4 \Rightarrow EI \, \delta_D = M_C (L/4)^3/6L - M_C (L/4)/24 \Rightarrow \delta_D = -M_C L^2/128EI$$

$$\text{At E, } x = 3L/4 \Rightarrow EI \, \delta_E = M_C (3L/4)^3/6L - M_C (L/4)^2/2 - M_C L(3L/4)/24 \Rightarrow \delta_E = M_C L^2/128EI$$



1.5.2 Distributed Load Shorter Than Span

Consider the loading scenarios shown below.

The corresponding bending moment expressions are written as: -

Case 1: $M = V_A x - \omega x^2/2$

Case 2: $M = V_A x - \omega[x - a]^2/2$

Case 3: $M = V_A x - \omega x^2/2$ } **These two are**
 Case 4: $M = V_A x - \omega[x - a]^2/2$ } **NOT correct!**

Notice that: -

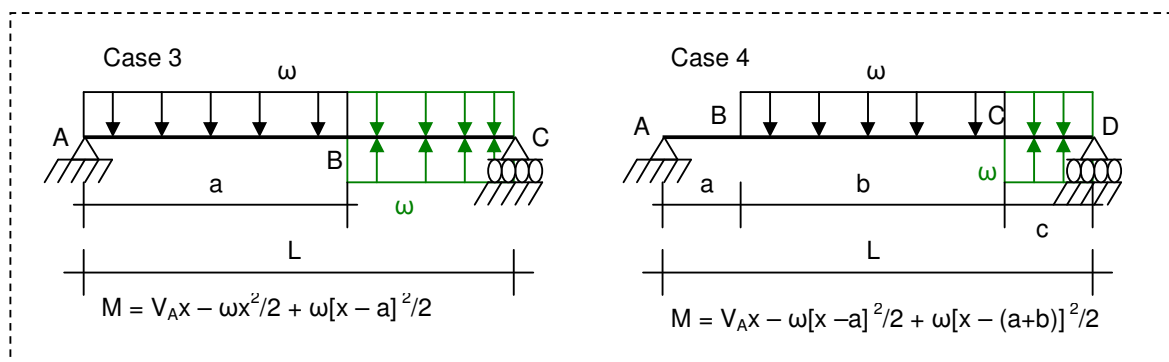
1. The moment expressions for case 1 and case 2 are correct and valid throughout the beam.
2. The moment expression for case 3 holds valid only in the region AB
3. The moment expression for case 4 holds valid in the regions AB and BC only.

The moment expressions in cases 3 and 4 can be valid only if the UDL extended to the right hand end of the beam.

Conclusions:

1. Macaulay's method is readily applicable when a UDL ends at the RH end of the beam, irrespective of where it starts.
2. When the UDL does not extend to the RH end of the beam, Macaulay's method appears to be handicapped. In order to remove this handicap, **the loading is made to extend to the right hand end of the beam**, then the loading which was not there is conveniently cancelled off by an equal but opposite loading, imagined to be applied from underneath the beam. **This also applies to a linearly distributed load shorter than span.**

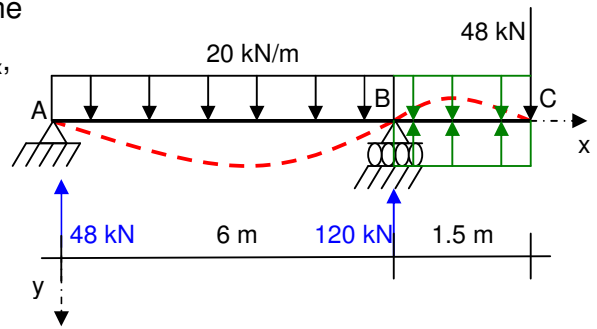
Case 3 and case 4 may thus be analysed as shown below.



Example 6

Determine the deflection of the free end C, δ_C , the slope at B, θ_B , and the maximum deflection, $\delta_{\max}$, induced in the span AB, stating where it occurs.

Assume $EI = \text{constant}$.



Solution

$$M = 48x - 20x^2/2 + 120[x - 6] + 20[x - 6]^2/2$$

$$EI \frac{d^2y}{dx^2} = -48x + 10x^2 - 120[x - 6] - 10[x - 6]^2$$

$$EI \frac{dy}{dx} = -24x^2 + 10x^3/3 - 60[x - 6]^2 - 10[x - 6]^3/3 + C_1$$

$$EI y = -8x^3 + 5x^4/6 - 20[x - 6]^3 - 5[x - 6]^4/6 + C_1x + C_2$$

Boundary conditions

$$1. y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. y|_6 = 0 \Rightarrow -8(6)^3 + (5/6)(6)^4 - 0 - 0 + 6C_1 = 0 \Rightarrow C_1 = 108$$

$$\text{Deflection profile: } EI \delta = -8x^3 + 5x^4/6 - 20[x - 6]^3 - 5[x - 6]^4/6 + 108x$$

$$\text{Slope profile: } EI \theta = -24x^2 + 10x^3/3 - 60[x - 6]^2 - 10[x - 6]^3/3 + 108$$

At the free end C: $x = 7.5$ m:

$$EI \delta_C = -8(7.5)^3 + (5/6)(7.5)^4 - 20(1.5)^3 - (5/6)(1.5)^4 + 108(7.5) = 0 \Rightarrow \delta_C = 0$$

$$\text{At B, } x = 6 \text{ m: } EI \theta_B = -24(6)^2 + 10(6)^3/3 + 108 = -36 \Rightarrow \theta_B = -36/EI$$

Within span AB, $x < 6$ m, max deflection occurs when the slope = 0

$$EI \frac{dy}{dx} = -24x^2 + 10x^3/3 + 108 = 0; \text{ Solving (by trial and error), } x \approx 2.676 \text{ m}$$

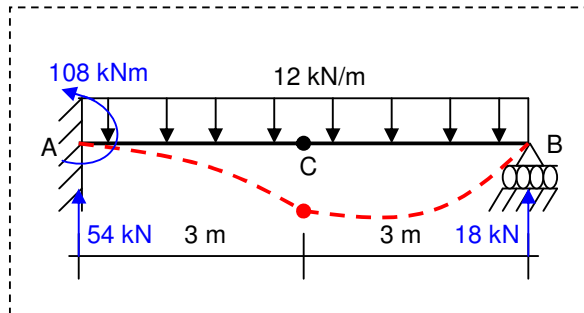
$$\text{Hence } EI \delta_{\max} \approx -8(2.676)^3 + (5/6)(2.676)^4 + 108(2.676) \Rightarrow \delta_{\max} \approx 178.439/EI$$

1.5.3 Internal Hinge

Consider the beam shown below.

Suppose it was required to find

δ_C and θ_B .



According to Macaulay's method, and assuming EI is constant:

$$M = 54x - 12x^2/2 - 108$$

$$EI \frac{d^2y}{dx^2} = -54x + 6x^2 + 108$$

$$EI \frac{dy}{dx} = -27x^2 + 2x^3 + 108x + C_1$$

$$EI y = -9x^3 + x^4/2 + 54x^2 + C_1x + C_2$$

Boundary conditions

$$1. \frac{dy}{dx} \Big|_0 = 0 \Rightarrow C_1 = 0$$

$$2. y \Big|_0 = 0 \Rightarrow C_2 = 0$$

Deflection profile: $EI \delta = -9x^3 + x^4/2 + 54x^2$

Slope profile: $EI \theta = -27x^2 + 2x^3 + 108x$

The deflections and slopes calculated using the above profiles are shown below.

x (m)	0	1	2	3	4	5	6
EI δ	0	91/2	152	567/2	416	1075/2	648
EI θ	0	83	124	135	128	115	108

These values of deflections and slopes show that: -

1. The deflection increases to 567/2EI at C, then **increases to 648/EI at B.**
2. The slope increases to 135/EI at C, then **remains positive up to B.**

The phrases in **red** do not make sense.

Conclusions:

1. The above profile equations hold only in region AC, i.e. **up to the internal hinge.**
2. The deflected shape has a **second order discontinuity (kink)** at the internal hinge C.
3. The internal hinge facilitates a **point of contra flexure** by allowing **free rotation.**
4. The above equations are therefore valid only in region AC.
5. The profile equations are therefore written with a proviso as shown below.

Deflection profile: $EI \delta = -9x^3 + x^4/2 + 54x^2$

Slope profile: $EI \theta = -27x^2 + 2x^3 + 108x$

For $0 \leq x \leq 3$

x measured from A

At C: $x = 3$; $EI \delta_C = -9(3)^3 + (1/2)(3)^4 + 54(3)^2 = 567/2 \Rightarrow \delta_C = 567/2EI$

In order to figure out the deflection and slope profiles in the region CB, we can work from C towards B. Region CB {x measured from C towards B}

$$M = 18x - 12x^2/2$$

$$EI \frac{d^2y}{dx^2} = -18x + 6x^2$$

$$EI \frac{dy}{dx} = -9x^2 + 2x^3 + C_3$$

$$EI y = -3x^3 + x^4/2 + C_3x + C_4$$

Boundary conditions

$$1. y|_0 = 567/2EI \Rightarrow C_4 = 567/2$$

$$2. y|_3 = 0 \Rightarrow C_3 = -81$$

$$\text{Deflection profile: } EI \delta = -3x^3 + x^4/2 - 81x + 567/2 \quad \left. \begin{array}{l} \text{For } 0 \leq x \leq 3 \\ x \text{ measured from C} \end{array} \right\}$$

$$\text{Slope profile: } EI \theta = -9x^2 + 2x^3 - 81$$

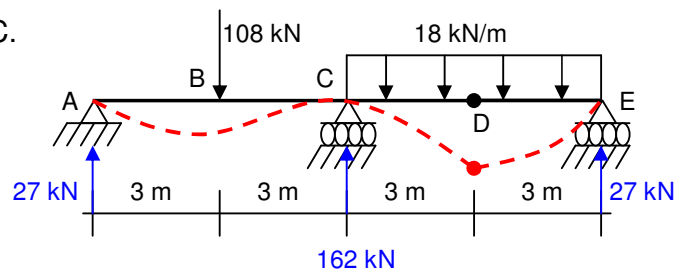
$$\text{At B: } x = 3; EI \theta_B = -9(3)^2 + 2(3)^3 - 81 = -108 \Rightarrow \theta_B = -108/EI$$

NOTE: The EI value does not necessarily have to be the same for the two regions

Example 8

Evaluate δ_D , θ_E and $\delta_{\max}$ within span AC.

Assume EI = constant.



Solution

Within the portion ABCD, with x measured from A:

$$M = 27x - 108[x - 3] + 162[x - 6] - 18[x - 6]^2/2$$

$$EI \frac{d^2y}{dx^2} = -27x + 108[x - 3] - 162[x - 6] + 9[x - 6]^2$$

$$EI \frac{dy}{dx} = -27x^2/2 + 54[x - 3]^2 - 81[x - 6]^2 + 3[x - 6]^3 + C_1$$

$$EI y = -9x^3/2 + 18[x - 3]^3 - 27[x - 6]^3 + 3[x - 6]^4/4 + C_1x + C_2$$

Boundary conditions

$$1. y|_0 = 0 \Rightarrow C_2 = 0$$

$$2. y|_6 = 0 \Rightarrow -9(6)^3/2 + 18(3)^3 + C_1(6) = 0 \Rightarrow C_1 = 81$$

$$\text{Defln profile: } EI \delta = -9x^3/2 + 18[x - 3]^3 - 27[x - 6]^3 + 3[x - 6]^4/4 + 81x \quad \left. \begin{array}{l} \text{For } 0 \leq x \leq 9 \\ x \text{ measured from A} \end{array} \right\}$$

$$\text{Slope profile: } EI \theta = -27x^2/2 + 54[x - 3]^2 - 81[x - 6]^2 + 3[x - 6]^3 + 81$$

$$\text{At D, } x = 9: \Rightarrow EI \delta_D = -9(9)^3/2 + 18(6)^3 - 27(3)^3 + 3(3)^4/4 + 81(9)$$

$$\Rightarrow \delta_D = 2673/4EI$$

$$\delta_{\max} \text{ within AC occurs when } dy/dx = 0 \Rightarrow -27x^2/2 + 54(x - 3)^2 + 81 = 0 \Rightarrow x \approx 2.586\text{m}$$

$$EI \delta_{\max} \approx -9(2.586)^3/2 + 81(2.586) \Rightarrow \delta_{\max} \approx 131.645/EI$$

Within the portion DE, with x measured from D

$$M = 27x - 18x^2/2$$

$$EI \, d^2y/dx^2 = -27x + 9x^2$$

$$EI \, dy/dx = -27x^2/2 + 3x^3 + C_3$$

$$EI \, y = -9x^3/2 + 3x^4/4 + C_3x + C_4$$

Boundary conditions

$$1. \, y|_0 = 2673/4EI \Rightarrow C_4 = 2673/4$$

$$2. \, y|_3 = 0 \Rightarrow C_3 = -405/2$$

$$\text{Deflection profile: } EI \, \delta = -9x^3/2 + 3x^4/4 - 405x/2 \quad \left. \vphantom{EI \, \delta} \right\} \text{ For } 0 \leq x \leq 3$$

$$\text{Slope profile: } EI \, \theta = -27x^2/2 + 3x^3 - 405/2 \quad \left. \vphantom{EI \, \theta} \right\} \text{ with x measured from E}$$

$$\text{At E, } x = 3 \Rightarrow EI \, \theta_E = -(27/2)(3)^2 + 3(3)^3 - 405/2 = -243 \Rightarrow \theta_E = -243/EI$$

Most Powerful Method
So long as $EI = \text{Constant}$